To prove: $P(k+1)$ is true.

$\therefore\ 1+3+5+\ldots+(2k-1)+(2k+1)$

$= P(k)+(2k+1)$

$= k^2+2k+1$

$= (k+1)^2$

$P(k+1)$ is true whenever $P(k)$ is true.

$\therefore\ P(n)$ is true for all $n \in N$.

Example 2.36

By Mathematical Induction, prove that $1^2+2^2+3^2+\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$, for all $n \in N$.

Solution

Let $P(n)$ denote the statement:

$$1^2+2^2+3^2+\ldots+n^2 = \frac{n(n+1)(2n+1)}{6}$$

Put $n = 1$

$\therefore$ LHS $= 1^2$

$= 1$

RHS $= \frac{1(1+1)(2+1)}{6}$

$= 1$

LHS = RHS

$\therefore\ P(1)$ is true.

Assume that $P(k)$ is true.

$P(k) = 1^2+2^2+3^2+\ldots+k^2$

$= \frac{k(k+1)(2k+1)}{6}$

Now,

$1^2+2^2+3^2+\ldots+k^2+(k+1)^2 = p(k)+(k+1)^2$

$= \frac{k(k+1)(2k+1)}{6}+(k+1)^2$

$= \frac{k(k+1)(2k+1)+6(k+1)^2}{6}$

$= \frac{(k+1)[k(2k+1)+6(k+1)]}{6}$

$= \frac{(k+1)(2k^2+7k+6)}{6}$

$= \frac{(k+1)(k+2)(2k+3)}{6}$

$\therefore\ P(k+1)$ is true whenever $P(k)$ is true

$\therefore\ P(n)$ is true for all $n \in N$.

Example 2.37

Show by the principle of mathematical induction that $2^{3n}-1$ is a divisible by 7, for all $n \in N$.

Solution

Let the given statement $P(n)$ be defined as $P(n) = 2^{3n}-1$

Step 1: Put $n = 1$

$\therefore\ P(1) = 2^3-1$

$= 7$ is divisible by 7

i.e., $P(1)$ is true.

Step 2: Let us assume that the statement is true for $n = k$ i.e., $P(k)$ is true.

We assume $2^{3k}-1$ is divisible by 7

$\Rightarrow\ 2^{3k}-1 = 7m$

Step 3: To prove that $P(k+1)$ is true

$P(k+1) = 2^{3(k+1)}-1$

$= 2^{(3k+3)}-1$

$= 2^{3k}\cdot 2^3-1$

$= 2^{3k}\cdot 8-1$

$= 2^{3k}\cdot(7+1)-1$

$= 2^{3k}\cdot 7+2^{3k}-1$

$= 2^{3k}\cdot 7+7m = 7(2^{3k}+m)$

which is divisible by 7

$P(k+1)$ is true whenever $p(k)$ is true

$\therefore\ P(n)$ is true for all $n \in N$.

Hence the proof.

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